

Pizza_sales_Analysis

June 20, 2026

1 Pizza_Sales_Analysis_Project

1.1 Overview:

- This project analyzes transactional sales data from a pizza store to uncover key insights and business trends.
- The dataset includes details such as order ID, pizza name, category, size, quantity, price, and timestamps.

1.2 Objectives:

- Calculate total revenue, total pizzas sold, and total number of orders
- Analyze sales by pizza category and size
- Identify daily, hourly, and monthly sales trends
- Highlight best-selling and least-selling pizzas
- Measure customer behavior using Average Order Value (AOV) and Average Pizzas per Order
- Provide visualizations for better decision-making

2 Import Libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

3 Importing Dataset

```
[4]: df = pd.read_csv('c:/Users/hp/Downloads/pizza_sales (1).csv')
df.head()
```

```
[4]:
```

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	\
0	1	1	hawaiian_m	1	01-01-2015	11:38:36	
1	2	2	classic_dlx_m	1	01-01-2015	11:57:40	
2	3	2	five_cheese_l	1	01-01-2015	11:57:40	
3	4	2	ital_supr_l	1	01-01-2015	11:57:40	
4	5	2	mexicana_m	1	01-01-2015	11:57:40	

	unit_price	total_price	pizza_size	pizza_category	\
0	13.25	13.25	M	Classic	
1	16.00	16.00	M	Classic	
2	18.50	18.50	L	Veggie	
3	20.75	20.75	L	Supreme	
4	16.00	16.00	M	Veggie	

	pizza_ingredients	\
0	Sliced Ham, Pineapple, Mozzarella Cheese	
1	Pepperoni, Mushrooms, Red Onions, Red Peppers,...	
2	Mozzarella Cheese, Provolone Cheese, Smoked Go...	
3	Calabrese Salami, Capocollo, Tomatoes, Red Oni...	
4	Tomatoes, Red Peppers, Jalapeno Peppers, Red O...	

	pizza_name
0	The Hawaiian Pizza
1	The Classic Deluxe Pizza
2	The Five Cheese Pizza
3	The Italian Supreme Pizza
4	The Mexicana Pizza

```
[5]: df.tail()
```

```
[5]:
```

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	\
48615	48616	21348	ckn_alfredo_m	1	31-12-2015	21:23:10	
48616	48617	21348	four_cheese_l	1	31-12-2015	21:23:10	
48617	48618	21348	napolitana_s	1	31-12-2015	21:23:10	
48618	48619	21349	mexicana_l	1	31-12-2015	22:09:54	
48619	48620	21350	bbq_ckn_s	1	31-12-2015	23:02:05	

	unit_price	total_price	pizza_size	pizza_category	\
48615	16.75	16.75	M	Chicken	
48616	17.95	17.95	L	Veggie	
48617	12.00	12.00	S	Classic	
48618	20.25	20.25	L	Veggie	
48619	12.75	12.75	S	Chicken	

	pizza_ingredients	\
48615	Chicken, Red Onions, Red Peppers, Mushrooms, A...	
48616	Ricotta Cheese, Gorgonzola Piccante Cheese, Mo...	
48617	Tomatoes, Anchovies, Green Olives, Red Onions,...	
48618	Tomatoes, Red Peppers, Jalapeno Peppers, Red O...	
48619	Barbecued Chicken, Red Peppers, Green Peppers,...	

	pizza_name
48615	The Chicken Alfredo Pizza
48616	The Four Cheese Pizza

```
48617      The Napolitana Pizza
48618      The Mexicana Pizza
48619  The Barbecue Chicken Pizza
```

4 *Understand The Data*

```
[6]: print(df.columns)
```

```
Index(['pizza_id', 'order_id', 'pizza_name_id', 'quantity', 'order_date',
      'order_time', 'unit_price', 'total_price', 'pizza_size',
      'pizza_category', 'pizza_ingredients', 'pizza_name'],
      dtype='object')
```

```
[7]: df.shape
```

```
[7]: (48620, 12)
```

```
[8]: # for rows
      print("The Rows of The Dataset: ",df.shape[0])
```

```
The Rows of The Dataset: 48620
```

```
[9]: print("The columns of The Dataset: ",df.shape[1])
```

```
The columns of The Dataset: 12
```

```
[10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48620 entries, 0 to 48619
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   pizza_id              48620 non-null  int64
1   order_id              48620 non-null  int64
2   pizza_name_id         48620 non-null  object
3   quantity              48620 non-null  int64
4   order_date            48620 non-null  object
5   order_time            48620 non-null  object
6   unit_price            48620 non-null  float64
7   total_price           48620 non-null  float64
8   pizza_size            48620 non-null  object
9   pizza_category        48620 non-null  object
10  pizza_ingredients     48620 non-null  object
11  pizza_name            48620 non-null  object
dtypes: float64(2), int64(3), object(7)
memory usage: 4.5+ MB
```

```
[11]: df.info
```

```
[11]: <bound method DataFrame.info of
quantity  order_date order_time \
0          1          1    hawaiian_m      1  01-01-2015   11:38:36
1          2          2   classic_dlx_m      1  01-01-2015   11:57:40
2          3          2   five_cheese_l      1  01-01-2015   11:57:40
3          4          2    ital_supr_l      1  01-01-2015   11:57:40
4          5          2    mexicana_m      1  01-01-2015   11:57:40
...
48615      48616      21348   ckn_alfredo_m      1  31-12-2015   21:23:10
48616      48617      21348   four_cheese_l      1  31-12-2015   21:23:10
48617      48618      21348   napolitana_s      1  31-12-2015   21:23:10
48618      48619      21349    mexicana_l      1  31-12-2015   22:09:54
48619      48620      21350    bbq_ckn_s      1  31-12-2015   23:02:05

unit_price  total_price  pizza_size  pizza_category \
0          13.25         13.25         M         Classic
1          16.00         16.00         M         Classic
2          18.50         18.50         L         Veggie
3          20.75         20.75         L         Supreme
4          16.00         16.00         M         Veggie
...
48615      16.75         16.75         M         Chicken
48616      17.95         17.95         L         Veggie
48617      12.00         12.00         S         Classic
48618      20.25         20.25         L         Veggie
48619      12.75         12.75         S         Chicken

pizza_ingredients \
0          Sliced Ham, Pineapple, Mozzarella Cheese
1  Pepperoni, Mushrooms, Red Onions, Red Peppers,...
2  Mozzarella Cheese, Provolone Cheese, Smoked Go...
3  Calabrese Salami, Capocollo, Tomatoes, Red Oni...
4  Tomatoes, Red Peppers, Jalapeno Peppers, Red O...
...
48615  Chicken, Red Onions, Red Peppers, Mushrooms, A...
48616  Ricotta Cheese, Gorgonzola Piccante Cheese, Mo...
48617  Tomatoes, Anchovies, Green Olives, Red Onions,...
48618  Tomatoes, Red Peppers, Jalapeno Peppers, Red O...
48619  Barbecued Chicken, Red Peppers, Green Peppers,...

pizza_name
0          The Hawaiian Pizza
1    The Classic Deluxe Pizza
2    The Five Cheese Pizza
3    The Italian Supreme Pizza
```

```

4           The Mexicana Pizza
...
48615  The Chicken Alfredo Pizza
48616      The Four Cheese Pizza
48617      The Napolitana Pizza
48618      The Mexicana Pizza
48619  The Barbecue Chicken Pizza

```

```
[48620 rows x 12 columns]>
```

5 *Data__Types*

```
[12]: # Data_Types
df.dtypes
```

```
[12]: pizza_id          int64
order_id          int64
pizza_name_id      object
quantity          int64
order_date        object
order_time        object
unit_price        float64
total_price       float64
pizza_size        object
pizza_category     object
pizza_ingredients  object
pizza_name        object
dtype: object
```

```
[13]: df.describe()
```

```
[13]:
```

	pizza_id	order_id	quantity	unit_price	total_price
count	48620.000000	48620.000000	48620.000000	48620.000000	48620.000000
mean	24310.500000	10701.479761	1.019622	16.494132	16.821474
std	14035.529381	6180.119770	0.143077	3.621789	4.437398
min	1.000000	1.000000	1.000000	9.750000	9.750000
25%	12155.750000	5337.000000	1.000000	12.750000	12.750000
50%	24310.500000	10682.500000	1.000000	16.500000	16.500000
75%	36465.250000	16100.000000	1.000000	20.250000	20.500000
max	48620.000000	21350.000000	4.000000	35.950000	83.000000

6 *Check_Missing_Values*

```
[14]: df.isnull().sum()
```

```
[14]: pizza_id          0
      order_id        0
      pizza_name_id   0
      quantity        0
      order_date       0
      order_time       0
      unit_price       0
      total_price      0
      pizza_size       0
      pizza_category   0
      pizza_ingredients 0
      pizza_name       0
      dtype: int64
```

```
[15]: df.isnull().sum().sum()
```

```
[15]: np.int64(0)
```

7 *Check_Duplicated_Values*

```
[16]: df.duplicated().sum()
```

```
[16]: np.int64(0)
```

8 *Key_Performance_Indicators*

- Total Revenue
- Total Pizzas Sold
- Total Orders
- Average Order Value (AOV)
- Average Pizza per Order

```
[17]: Total_revenue = df['total_price'].sum()
      Total_Pizzas_Sold = df['quantity'].sum()
      Total_Orders = df['order_id'].nunique()
      Average_Order_Value = Total_revenue / Total_Orders
      Average_Pizza_per_Order = Total_Pizzas_Sold / Total_Orders
```

```
[18]: print(f"Total Revenue :           ${Total_revenue:,.2f}")
      print(f"Total Pizzas Sold :       {Total_Pizzas_Sold:,}")
      print(f"Total Orders :           {Total_Orders:,.2f}")
      print(f"Average Order Value :     ${Average_Order_Value:,.2f}")
```

```
print(f"Average Pizza per Order:      {Average_Pizza_per_Order:,.2f}")
```

```
Total Revenue :           $817,860.05
Total Pizzas Sold :         49,574
Total Orders :             21,350.00
Average Order Value  :      $38.31
Average Pizza per Order:    2.32
```

9 *Data Visualization*

- Ingredient Analysis

```
[37]: ingredients = (
      df['pizza_ingredients']
      .str.split(',')
      .explode().str.strip()
      .value_counts()
      .reset_index()
      .rename(columns={'index': 'count', 'pizza_ingredients': 'ingredients'})
      )
```

```
[38]: print(ingredients)
```

	ingredients	count
0	Garlic	27422
1	Tomatoes	26601
2	Red Onions	19547
3	Red Peppers	16284
4	Mozzarella Cheese	10333
..	...	...
60	Brie Carre Cheese	480
61	Prosciutto	480
62	Caramelized Onions	480
63	Pears	480
64	Thyme	480

[65 rows x 2 columns]

```
[36]: ingredients.head(10)
```

```
[36]:
```

	ingredients	count
0	Garlic	27422
1	Tomatoes	26601
2	Red Onions	19547
3	Red Peppers	16284
4	Mozzarella Cheese	10333
5	Pepperoni	10300

6	Spinach	10012
7	Mushrooms	9624
8	Chicken	8443
9	Capocollo	6572

10 Trends

- Daily_Trend -> Total Orders

```
[41]: df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)
df['day_name'] = df['order_date'].dt.day_name()

weekday_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday",
                 ↪ "Saturday", "Sunday"]
df['day_name'] = pd.Categorical(df['day_name'], categories=weekday_order,
                               ↪ ordered=True)

orders_by_day = df.groupby('day_name', observed=False)['order_id'].nunique()
orders_by_day = orders_by_day.sort_index()
```

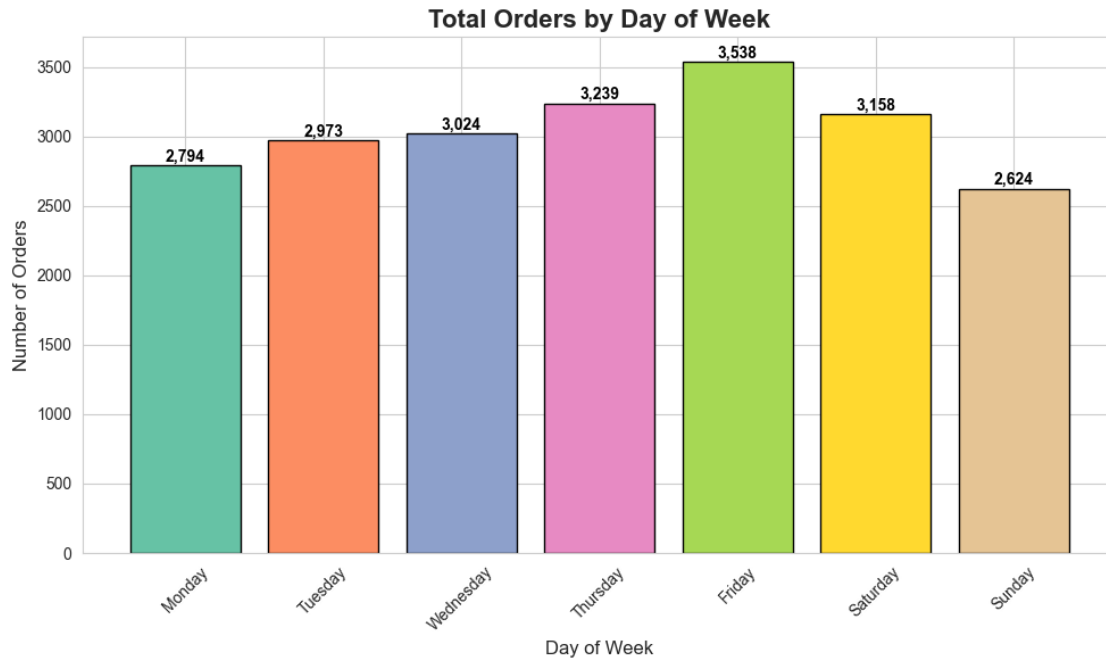
```
[42]: sns.set_style("whitegrid")
palette = sns.color_palette("Set2")

# Create plot
fig, ax = plt.subplots(figsize=(10, 6))
bars = ax.bar(orders_by_day.index, orders_by_day.values, color=palette,
              ↪ edgecolor='black')

plt.title("Total Orders by Day of Week", fontsize=16, fontweight='bold')
plt.xlabel("Day of Week", fontsize=12)
plt.ylabel("Number of Orders", fontsize=12)
plt.xticks(rotation=45, fontsize=10)
plt.yticks(fontsize=10)

for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, height + 5, f'{height:,}',
            ha='center', va='bottom', fontsize=10, fontweight='bold',
            ↪ color='black')

plt.tight_layout()
plt.show()
```

- Daily_Trend -> Total Revenue

```
[55]: df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)

df['day_name'] = df['order_date'].dt.day_name()

weekday_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday",
                 ↪ "Saturday", "Sunday"]

df['day_name'] = pd.Categorical(df['day_name'], categories=weekday_order,
                               ↪ ordered=True)

orders_by_day = df.groupby('day_name', observed=False)['total_price'].sum()

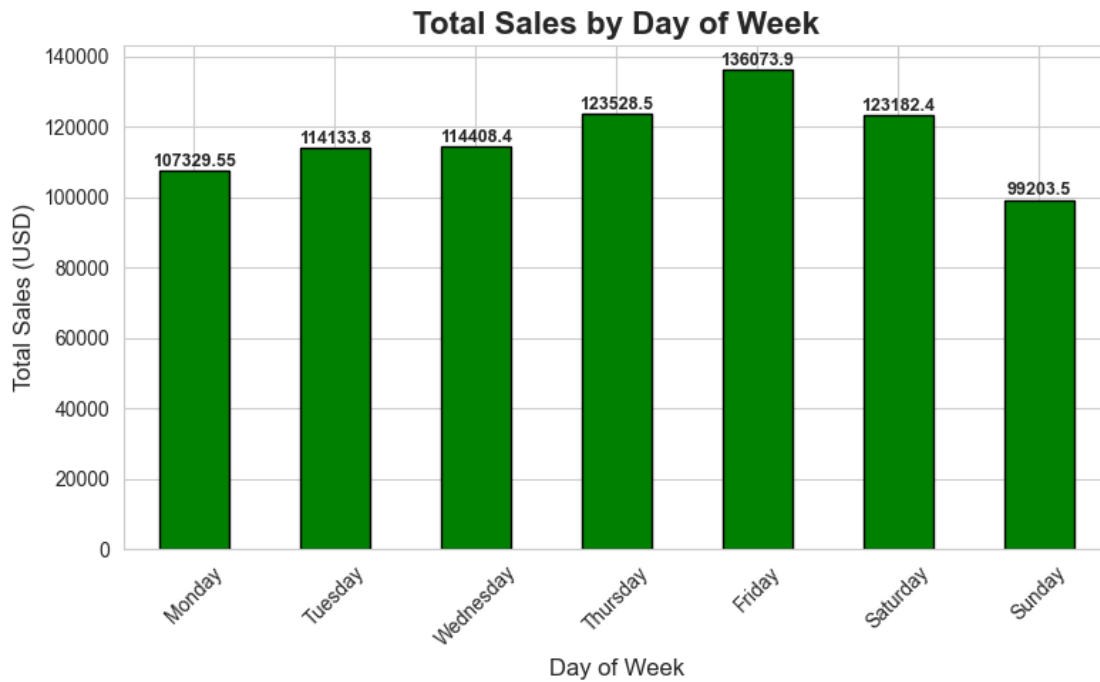
ax = orders_by_day.plot(kind='bar', figsize=(8,5), color="green",
                        ↪ edgecolor='black')

plt.title("Total Sales by Day of Week", fontsize=16, fontweight='bold')
plt.xlabel("Day of Week", fontsize=12)
plt.ylabel("Total Sales (USD)", fontsize=12)
plt.xticks(rotation=45, fontsize=10)
plt.yticks(fontsize=10)

for i, val in enumerate(orders_by_day):
```

```
plt.text(i, val + 20, str(val), ha='center', va='bottom', fontsize=9,
↪fontweight='bold')
```

```
plt.tight_layout()
plt.show()
```



11 *Hourelly_Trend*

```
[60]: df['order_time'] = pd.to_datetime(df['order_time'], format='%H:%M:%S')

df['order_hour'] = df['order_time'].dt.hour

orders_by_hour = df.groupby('order_hour', observed=False)['order_id'].nunique()

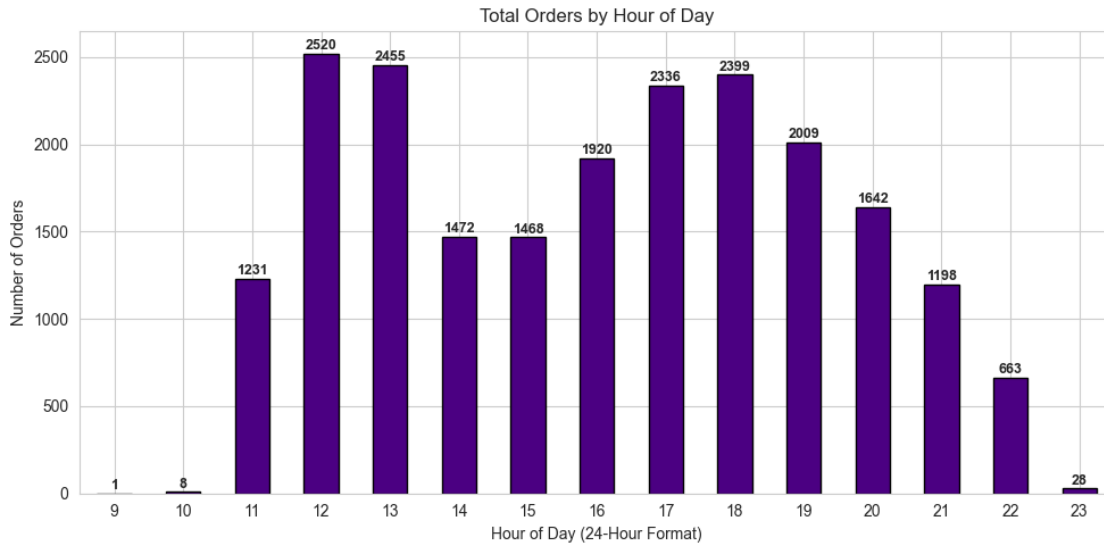
ax = orders_by_hour.plot(kind='bar', figsize=(10,5), color='Indigo',
↪edgecolor='black')

plt.title("Total Orders by Hour of Day")
plt.xlabel("Hour of Day (24-Hour Format)")
plt.ylabel("Number of Orders")
plt.xticks(rotation=0)

for i, val in enumerate(orders_by_hour):
```

```
plt.text(i, val + 5, str(val), ha='center', va='bottom', fontsize=9,
↪fontweight='bold')
```

```
plt.tight_layout()
plt.show()
```



12 Monthly_Trend

```
[70]: df['order_date'] = pd.to_datetime(df['order_date'], dayfirst=True)

df['month_name'] = df['order_date'].dt.month_name()

month_order = ["January", "February", "March", "April", "May", "June",
               "July", "August", "September", "October", "November", "December"]

df['month_name'] = pd.Categorical(df['month_name'], categories=month_order,
↪ordered=True)

orders_by_month = df.groupby('month_name', observed=False)['order_id'].nunique()

plt.figure(figsize=(10,5))
plt.fill_between(orders_by_month.index, orders_by_month.values, color="orange",
↪alpha=0.6)
plt.plot(orders_by_month.index, orders_by_month.values, color="green",
↪linewidth=2, marker='o')

plt.title("Total Orders by Month")
```

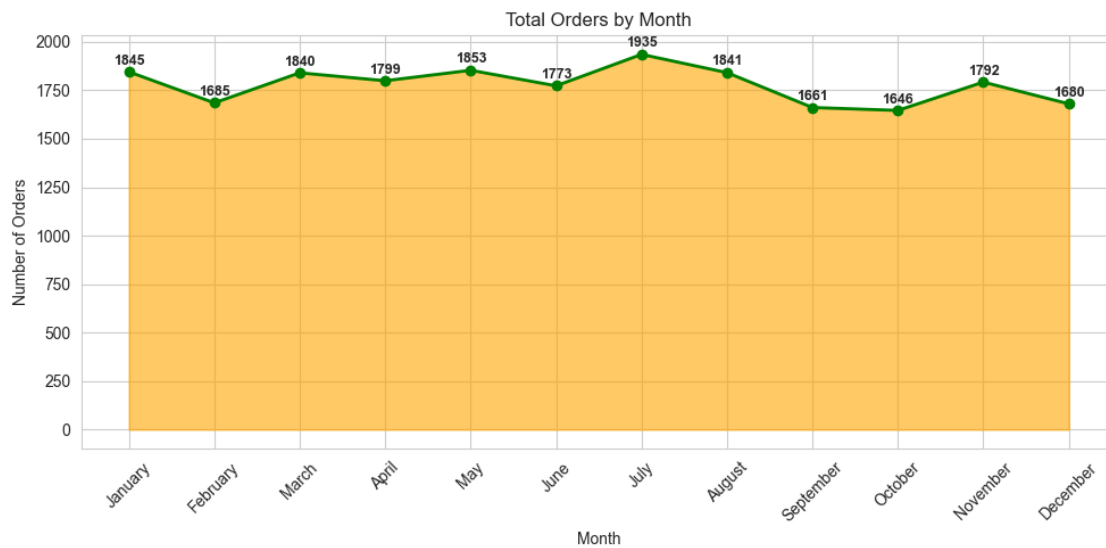
```

plt.xlabel("Month")
plt.ylabel("Number of Orders")
plt.xticks(rotation=45)

for i, val in enumerate(orders_by_month):
    plt.text(i, val + 20, str(val), ha='center', va='bottom', fontsize=9,
             fontweight='bold')

plt.tight_layout()
plt.show()

```



13 Sales_Category_Percentage

```

[79]: category_sales = df.groupby('pizza_category')['total_price'].sum()

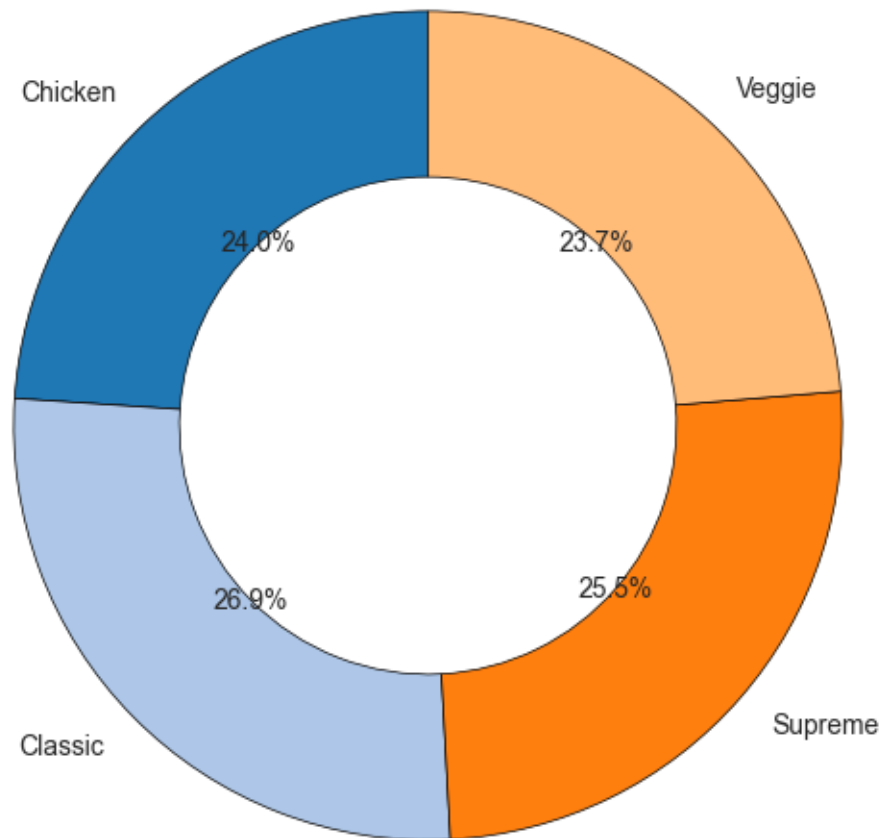
category_pct = category_sales / category_sales.sum() * 100

plt.figure(figsize=(7, 7))
colors = plt.get_cmap('tab20').colors
plt.pie(category_pct, labels=category_pct.index, autopct='%1.1f%%',
        startangle=90,
        colors=colors, wedgeprops={'edgecolor':'black', 'width':0.4})

plt.title("Percentage of Sales by Pizza Category")
plt.show()

```

Percentage of Sales by Pizza Category



14 *Heatmap*

```
[82]: print(plt.style.available)
```

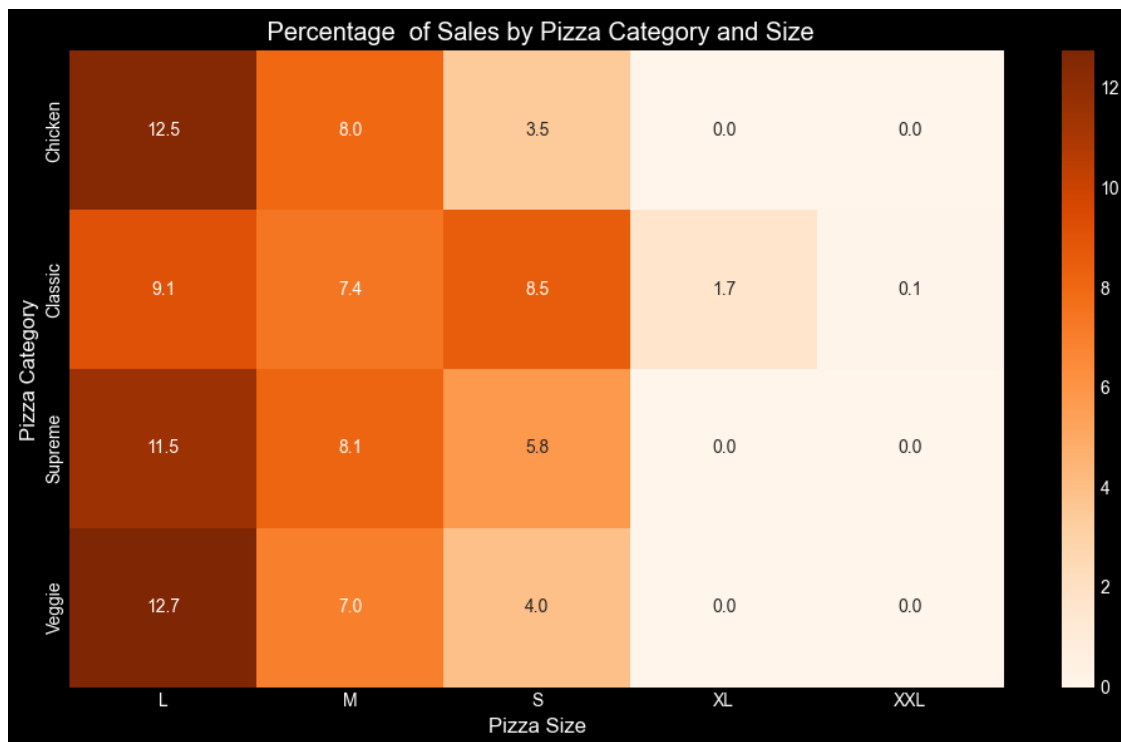
```
['Solarize_Light2', '_classic_test_patch', '_mpl-gallery', '_mpl-gallery-nogrid', 'bmh', 'classic', 'dark_background', 'fast', 'fivethirtyeight', 'ggplot', 'grayscale', 'seaborn-v0_8', 'seaborn-v0_8-bright', 'seaborn-v0_8-colorblind', 'seaborn-v0_8-dark', 'seaborn-v0_8-dark-palette', 'seaborn-v0_8-darkgrid', 'seaborn-v0_8-deep', 'seaborn-v0_8-muted', 'seaborn-v0_8-notebook', 'seaborn-v0_8-paper', 'seaborn-v0_8-pastel', 'seaborn-v0_8-poster', 'seaborn-v0_8-talk', 'seaborn-v0_8-ticks', 'seaborn-v0_8-white', 'seaborn-v0_8-whitegrid', 'tableau-colorblind10']
```

```
[ ]: sales_pivot = df.pivot_table(
    index='pizza_category',
    columns='pizza_size',
    values='total_price',
    aggfunc='sum',
    fill_value=0
)

sales_pct = sales_pivot / sales_pivot.sum().sum() * 100

plt.figure(figsize=(10, 6))
sns.heatmap(
    sales_pct,
    annot=True,
    fmt=".1f",
    cmap="Oranges",
    linecolor="white"
)

plt.title(" Percentage of Sales by Pizza Category and Size")
plt.xlabel("Pizza Size")
plt.ylabel("Pizza Category")
plt.tight_layout()
plt.show()
```



```
[96]: import matplotlib.pyplot as plt

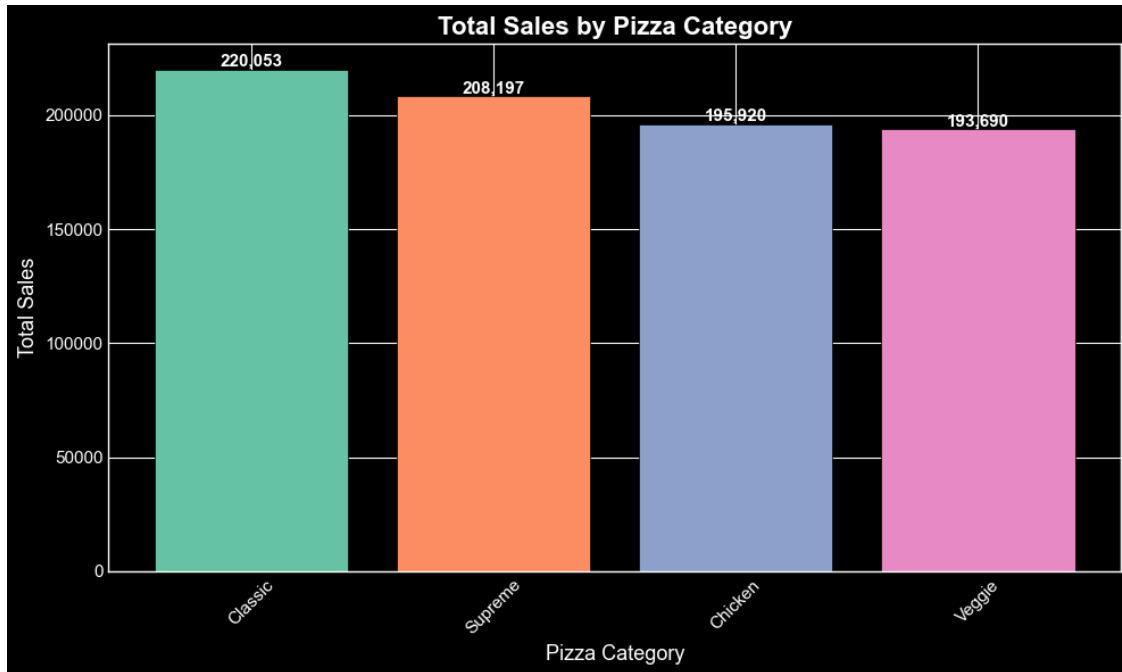
# Group total sales by pizza category
category_sales = df.groupby('pizza_category')['total_price'].sum().
    ↪sort_values(ascending=False)

# Create a color palette with unique colors
colors = plt.cm.Set2(range(len(category_sales)))

# Plot bar chart
plt.figure(figsize=(10, 6))
bars = plt.bar(category_sales.index, category_sales.values, color=colors,
    ↪edgecolor='black')

# Add value labels on top of each bar
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, height + 1, f'{height:,.0f}',
        ha='center', va='bottom', fontsize=11, fontweight='bold')

# Styling
plt.title("Total Sales by Pizza Category", fontsize=16, fontweight='bold')
plt.xlabel("Pizza Category", fontsize=13)
plt.ylabel("Total Sales", fontsize=13)
plt.xticks(rotation=45, fontsize=11)
plt.yticks(fontsize=11)
plt.tight_layout()
plt.show()
```



15 Top 10 Best-Selling Pizza Products by Total Sales

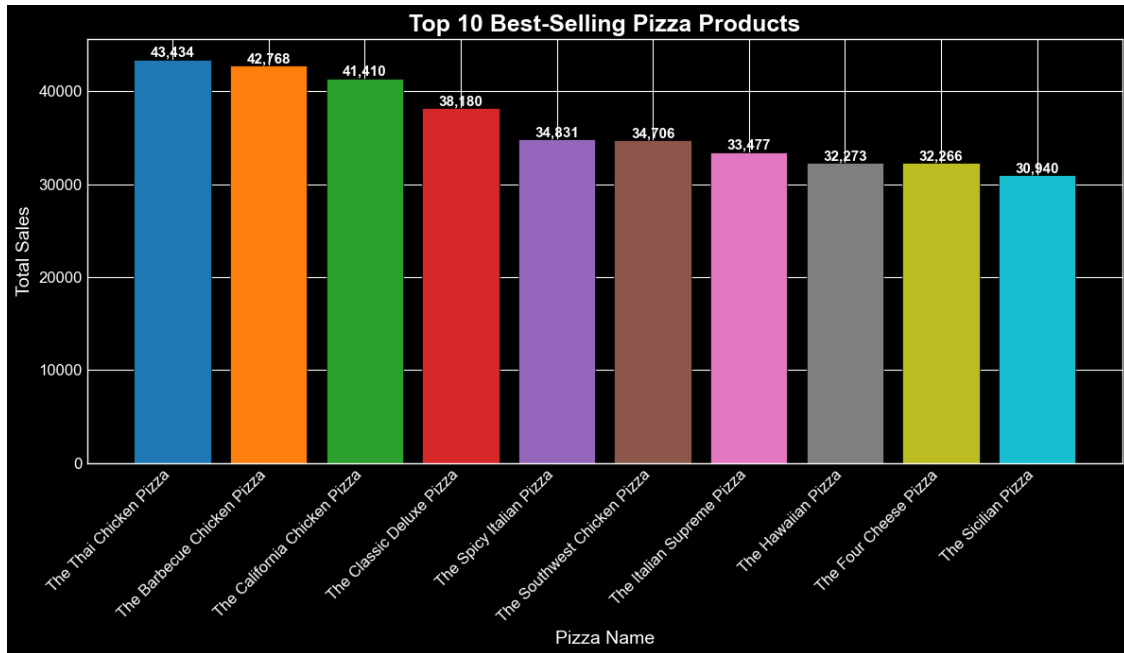
```
[99]: top_10 = df.groupby('pizza_name')['total_price'].sum().
      ↪sort_values(ascending=False).head(10)

colors = plt.cm.tab10(range(10))

plt.figure(figsize=(12, 7))
bars = plt.bar(top_10.index, top_10.values, color=colors, edgecolor='black')

for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width() / 2, height + 1,
             f'{height:,.0f}', ha='center', va='bottom', fontsize=11,
             ↪fontweight='bold')

plt.title("Top 10 Best-Selling Pizza Products", fontsize=18, fontweight='bold')
plt.xlabel("Pizza Name", fontsize=14)
plt.ylabel("Total Sales", fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=12)
plt.yticks(fontsize=12)
plt.tight_layout()
plt.show()
```

[]: